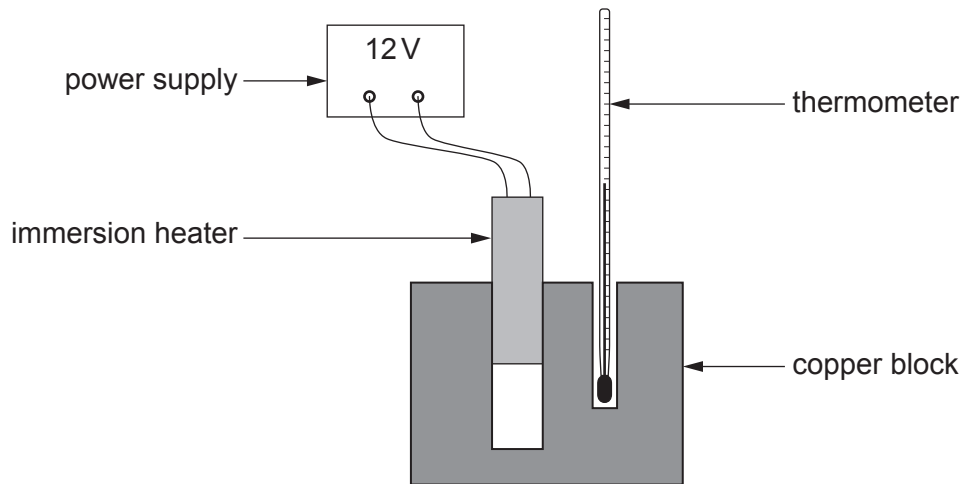


4. The apparatus used by a group of students to determine the specific heat capacity of copper is shown below.



- (a) (i) The immersion heater has a power of 50 W.
It heats the copper block for 150 s.
Use the equation:

$$\text{energy transferred} = \text{power} \times \text{time}$$

to calculate the energy transferred by the heater.

[2]

energy transferred = J

- (ii) The temperature of the copper block at the start of the experiment is 20 °C.
Its temperature after heating is 35 °C.
Calculate the change in temperature of the copper block.

[1]

change in temperature = °C



- (iii) The mass of the copper block is 1 kg.

Use the answers in parts (i) and (ii) and the equation:

$$\text{specific heat capacity} = \frac{\text{energy transferred}}{\text{mass} \times \text{change in temperature}}$$

to calculate the specific heat capacity of copper.

[2]

specific heat capacity, $c = \dots\dots\dots \text{J/kg } ^\circ\text{C}$

- (iv) Another group of students get a value for the specific heat capacity of $455 \text{ J/kg } ^\circ\text{C}$.
The true value of the specific heat capacity of copper is $385 \text{ J/kg } ^\circ\text{C}$.

- I. Calculate the difference between this group's value and the true value.

[1]

difference = $\dots\dots\dots \text{J/kg } ^\circ\text{C}$

- II. Use the equation:

$$\% \text{ difference} = \frac{\text{difference}}{\text{true value}} \times 100$$

to calculate the % difference between this group's value and the true value.

[2]

% difference = $\dots\dots\dots$



- (b) Tick (✓) the boxes next to **two** improvements that could be made to this experiment that would lead to a more accurate value for the specific heat capacity of copper. [2]

Use a heater with a larger power

☐

Heat the block for longer

☐

Wrap the block in insulating material

☐

Push the heater fully into the block

☐
☐

10

